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Home Intrusion Detection System

An Arduino-Based Smart Security System

COE 381 – Microprocessors

GROUP 4

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Contents

1 Introduction **1**
1.1 Motivation . . . 1
1.2 Objectives . . . 1

2 Background and Literature Review **1**
2.1 Microcontroller-Based Security Systems . . . 1
2.2 Sensor Technologies . . . 2
2.2.1 Ultrasonic Ranging (HC-SR04) . . . 2
2.2.2 Passive Infrared (PIR) Sensor . . . 2
2.2.3 4×4 Matrix Keypad . . . 2
2.2.4 I^2C LCD (LiquidCrystal_I2C) . . . 2

3 System Design **2**
3.1 Hardware Architecture . . . 2
3.2 Pin Assignment . . . 3
3.3 Finite-State Machine . . . 3
3.3.1 State Descriptions . . . 4

4 Implementation **4**
4.1 Software Overview . . . 4
4.2 Key Functions . . . 5
4.2.1 Ultrasonic Distance Measurement . . . 5
4.2.2 Non-Blocking Buzzer Alternation . . . 5
4.2.3 Non-Blocking RGB Cycling . . . 5
4.2.4 Dynamic I^2C Address Scan . . . 5
4.2.5 System Reset . . . 6
4.3 Main Loop Logic . . . 6

5 Full Source Code **6**

6 Results and Discussion **10**
6.1 System Behaviour Verification . . . 10
6.2 Non-Blocking Timing . . . 11
6.3 Sensor Fusion Effectiveness . . . 11
6.4 Limitations . . . 11

7 Conclusion **12**

References **12**

List of Figures

1 Finite-State Machine for the Home Intrusion Detection System 4

List of Tables

1 Bill of Materials 3
2 Arduino Mega Pin Assignments 3
3 Test Scenarios and Observed Outcomes 11

1 Introduction

Security remains a critical concern in residential and commercial environments. Traditional security systems often rely on passive measures such as locks and alarms that activate only after a breach has occurred. Modern embedded systems offer a more proactive approach, integrating multiple sensor modalities, intelligent state management, and real-time response mechanisms into a compact, low-cost platform.

This report presents the design, implementation, and evaluation of a **Home Intrusion Detection System** built on the Arduino Mega 2560 microcontroller platform. The system combines an ultrasonic proximity sensor, a passive infrared (PIR) motion sensor, a membrane keypad, an I^2C liquid-crystal display (LCD), an RGB LED, and a dual-buzzer alarm unit to deliver a layered security solution suitable for doorways, entry points, or any monitored perimeter.

1.1 Motivation

Commercially available security systems are expensive and often require professional installation. This project demonstrates that a functionally equivalent system can be constructed from widely available components, programmed using the Arduino framework, and deployed for a fraction of the cost. It also serves as a practical application of core microcontroller concepts covered in COE 381, including digital I/O, serial communication protocols (I^2C, UART), interrupt-free state machines, and real-time sensor fusion.

1.2 Objectives

The specific objectives of this project are to:

i. Design a finite-state machine (FSM) that governs system behaviour across five distinct operational states.

ii. Interface a 4×4 matrix keypad for PIN-based user authentication.

iii. Integrate an HC-SR04 ultrasonic sensor and a PIR sensor for proximity and motion detection respectively.

iv. Provide multi-modal feedback via an RGB LED, a 16×2 I^2C LCD, and a dual-buzzer alarm.

v. Implement non-blocking timing logic to allow concurrent sensor polling and output control.

2 Background and Literature Review

2.1 Microcontroller-Based Security Systems

The Arduino platform, introduced in 2005, has become a de facto standard for rapid prototyping of embedded systems in both academic and industrial settings. Its open-source hardware and software ecosystem, combined with a large library of peripheral drivers, makes it well-suited to security applications [1].

Previous work in this domain has demonstrated systems using single-sensor approaches (PIR-only or ultrasonic-only). The novelty of the present design lies in the sensor fusion

strategy: the ultrasonic sensor provides reliable, short-range proximity detection independent of ambient temperature, while the PIR sensor extends coverage to wider-angle thermal motion detection, collectively reducing false-negative rates.

2.2 Sensor Technologies

2.2.1 Ultrasonic Ranging (HC-SR04)

The HC-SR04 operates on the principle of time-of-flight measurement. A 10 μs trigger pulse causes the sensor to emit eight 40 kHz ultrasonic bursts. The echo pin remains HIGH for the duration of the round-trip travel time t. Distance d is then:

$$d = \frac{v_s \cdot t}{2} \tag{1}$$

where $v_s \approx 343\,\mathrm{m/s}$ is the speed of sound at room temperature, or equivalently $d \approx 0.034 \times t\,\mathrm{cm}$ when t is expressed in microseconds.

2.2.2 Passive Infrared (PIR) Sensor

PIR sensors detect changes in infrared radiation emitted by warm bodies within their field of view (typically 110° cone, up to 7 m range). They are highly sensitive to lateral movement and consume very little power, making them complementary to the directional ultrasonic sensor.

2.2.3 4×4 Matrix Keypad

The membrane keypad uses a row-column scanning technique. The Arduino drives each row LOW in turn and reads the column pins to identify which key, if any, is pressed. The `Keypad` library abstracts this into a simple `getKey()` call with built-in debouncing.

2.2.4 I²C LCD (LiquidCrystal_I2C)

The 16×2 LCD is driven through a PCF8574 I/O expander over the I²C bus (SDA on pin 20, SCL on pin 21 on the Arduino Mega), reducing the pin count from eight parallel data lines to just two. An I²C bus scan at startup dynamically resolves the LCD's address, improving portability across modules.

3 System Design

3.1 Hardware Architecture

Table 1 lists all hardware components used in the system.

Table 1: Bill of Materials

Component	Description	Quantity
Arduino Mega 2560	Main microcontroller (ATmega2560, 16 MHz)	1
HC-SR04	Ultrasonic distance sensor (2–400 cm)	1
PIR Sensor	Passive infrared motion detector	1
4×4 Keypad	Membrane matrix keypad	1
16×2 I^2C LCD	Character display with PCF8574 backpack	1
RGB LED (common cathode)	Red/Green/Blue status indicator	1
Active Buzzer	5 V buzzer for alert/alarm tones	2
Resistors (220 Ω)	Current-limiting for RGB LED	3
Breadboard & jumper wires	Prototyping connections	–
5 V DC supply	USB or external regulated supply	1

3.2 Pin Assignment

Table 2 summarises the Arduino Mega pin assignments.

Table 2: Arduino Mega Pin Assignments

Pin	Direction	Peripheral	Function
2	OUTPUT	Buzzer 1	Alert/alarm tone channel 1
3	OUTPUT	Buzzer 2	Alert/alarm tone channel 2
4	OUTPUT	RGB LED (Blue)	Blue state indicator
5	OUTPUT	RGB LED (Green)	Green state indicator
6	OUTPUT	RGB LED (Red)	Red state indicator
7	OUTPUT	HC-SR04 TRIG	Trigger pulse output
8	INPUT	HC-SR04 ECHO	Echo pulse input
9–12	OUTPUT	Keypad rows	Row drive lines
20	I^2C SDA	LCD	Serial data
21	I^2C SCL	LCD	Serial clock
22–25	INPUT	Keypad cols	Column read lines
43	INPUT	PIR sensor	Motion detection input

3.3 Finite-State Machine

The system behaviour is governed by a five-state FSM. Figure 1 illustrates the state transitions.

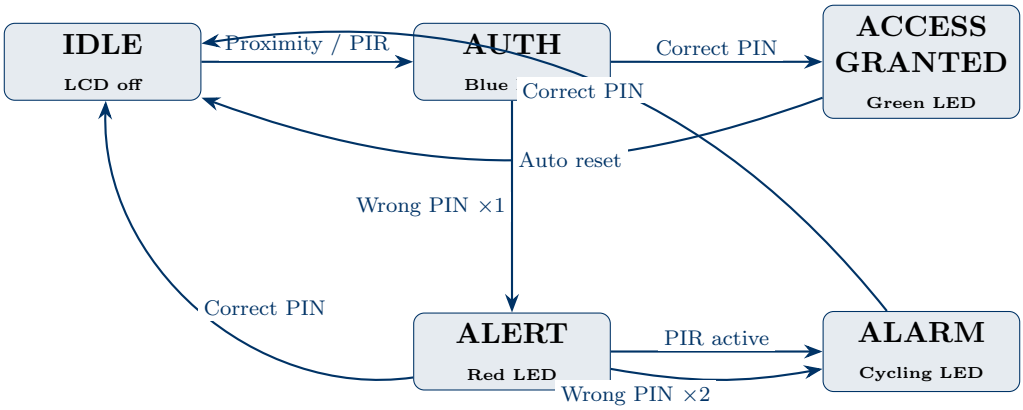

Figure 1: Finite-State Machine for the Home Intrusion Detection System

3.3.1 State Descriptions

IDLE

The system waits with the LCD backlight off and all outputs de-energised. The main loop continuously polls the ultrasonic sensor (threshold: 30 cm) and the PIR pin. Either trigger advances the FSM to AUTH.

AUTH (Authentication)

The LCD prompts the user for a four-digit PIN. The RGB LED glows blue. Keypad input is buffered character by character; '*' clears the buffer. A correct PIN transitions to ACCESS GRANTED; an incorrect PIN increments `wrongAttempts` and transitions to ALERT.

ACCESS GRANTED

The RGB LED turns green, the LCD displays "Accepted!", and after a 2-second hold the system resets to IDLE.

ALERT

One incorrect PIN has been entered. The LED flashes red at 300 ms intervals. A single beep was emitted on entry. The user may still enter the correct PIN to return to IDLE. A second incorrect attempt or PIR activation escalates to ALARM.

ALARM

The dual buzzers alternate at 4 kHz/1 s intervals. The RGB LED cycles through Red–Blue–Purple at 200 ms intervals. The LCD displays "ALARM!!!". Only a correct PIN silences the alarm and resets the system.

4 Implementation

4.1 Software Overview

The firmware is written in C++ using the Arduino framework and relies on two external libraries:

- `Keypad.h` – matrix keypad scanning and debouncing.
- `LiquidCrystal_I2C.h` – I^2C LCD control.

All timing-sensitive operations (LED cycling, buzzer alternation) use `millis()`-based non-blocking delays, ensuring that keypad scanning and sensor polling are never stalled by `delay()` calls in the alarm path.

4.2 Key Functions

4.2.1 Ultrasonic Distance Measurement

```
long getDistance() {
    digitalWrite(TRIG_PIN, LOW);  delayMicroseconds(2);
    digitalWrite(TRIG_PIN, HIGH); delayMicroseconds(10);
    digitalWrite(TRIG_PIN, LOW);
    long dur = pulseIn(ECHO_PIN, HIGH, 30000); // 30 ms timeout
    return dur * 0.034 / 2;                    // convert to cm
}
```

Listing 1: Ultrasonic distance measurement function

The 30 ms timeout in `pulseIn()` prevents indefinite blocking when no echo is received (e.g., target out of range).

4.2.2 Non-Blocking Buzzer Alternation

```
void buzzerAlternate() {
    if (millis() - lastBuzzerChange >= 1000) {
        buzzerPhase = (buzzerPhase + 1) % 2;
        lastBuzzerChange = millis();
        if (buzzerPhase == 0) {
            tone(BUZZER1, 4000); noTone(BUZZER2);
        } else {
            tone(BUZZER2, 4000); noTone(BUZZER1);
        }
    }
}
```

Listing 2: Non-blocking dual-buzzer alternation in alarm state

4.2.3 Non-Blocking RGB Cycling

```
void cycleRGB() {
    if (millis() - lastColorChange >= 200) {
        colorPhase = (colorPhase + 1) % 3;
        if (colorPhase == 0)      rgbRed();
        else if (colorPhase == 1) rgbBlue();
        else                      rgbPurple();
        lastColorChange = millis();
    }
}
```

Listing 3: Non-blocking RGB colour cycling during alarm

4.2.4 Dynamic I²C Address Scan

```
byte scanI2C() {
    for (byte addr = 1; addr < 127; addr++) {
        Wire.beginTransmission(addr);
```

```
        if (Wire.endTransmission() == 0) return addr;
    }
    return 0;
}
```

Listing 4: I2C bus scan for dynamic LCD address resolution

The scan iterates through all valid 7-bit I^2C addresses and returns the first responding device. This removes the hard-coded dependency on address `0x27`, improving compatibility across different LCD modules.

4.2.5 System Reset

```
void resetSystem() {
    wrongAttempts = 0;
    enteredPass   = "";
    systemArmed   = false;
    colorPhase    = 0;
    buzzerOff();
    rgbOff();
    lcd.clear();
    lcd.noBacklight();
}
```

Listing 5: Full system reset to IDLE state

4.3 Main Loop Logic

The `loop()` function implements the FSM without a formal switch-case construct, instead using a guard variable (`systemArmed`) and inline conditional blocks. The execution flow is summarised below:

1. **IDLE check:** If `!systemArmed`, poll ultrasonic and PIR. If triggered, set `systemArmed = true` and return.
2. **Alarm / Alert:** If `wrongAttempts >= 2` or PIR is active, handle alarm output and keypad input. Accept correct PIN to reset.
3. **Normal entry:** Read keypad; accumulate PIN. On four digits, validate against `PASSWORD`. Correct – reset; incorrect – increment `wrongAttempts` and beep.

5 Full Source Code

```
#include <Keypad.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

//        PINS


#define BUZZER1    2
#define BUZZER2    3
#define RGB_BLUE   4
#define RGB_GREEN  5
#define RGB_RED    6
```

```
#define TRIG_PIN   7
#define ECHO_PIN   8
#define PIR_PIN    43

// LCD
LiquidCrystal_I2C lcd(0x27, 16, 2);
byte lcdAddress = 0x27;

//        KEYPAD


const byte ROWS = 4, COLS = 4;
char keys[ROWS][COLS] = {
    {'1','2','3','A'}, {'4','5','6','B'},
    {'7','8','9','C'}, {'*','0','#','D'}
};
byte rowPins[ROWS] = {9, 10, 11, 12};
byte colPins[COLS] = {22, 23, 24, 25};
Keypad keypad = Keypad(makeKeymap(keys), rowPins, colPins, ROWS, COLS);

//        STATE


const String PASSWORD = "1234";
String enteredPass    = "";
int    wrongAttempts  = 0;
bool   systemArmed    = false;
int    colorPhase     = 0;
unsigned long lastColorChange  = 0;
int    buzzerPhase    = 0;
unsigned long lastBuzzerChange = 0;

//        RGB HELPERS


void rgbOff()    { digitalWrite(RGB_RED,0);digitalWrite(RGB_GREEN,0);
    digitalWrite(RGB_BLUE,0); }
void rgbRed()    { digitalWrite(RGB_RED,1);digitalWrite(RGB_GREEN,0);
    digitalWrite(RGB_BLUE,0); }
void rgbGreen()  { digitalWrite(RGB_RED,0);digitalWrite(RGB_GREEN,1);
    digitalWrite(RGB_BLUE,0); }
void rgbBlue()   { digitalWrite(RGB_RED,0);digitalWrite(RGB_GREEN,0);
    digitalWrite(RGB_BLUE,1); }
void rgbPurple() { digitalWrite(RGB_RED,1);digitalWrite(RGB_GREEN,0);
    digitalWrite(RGB_BLUE,1); }

void cycleRGB() {
    if (millis() - lastColorChange >= 200) {
        colorPhase = (colorPhase + 1) % 3;
        if      (colorPhase == 0) rgbRed();
        else if (colorPhase == 1) rgbBlue();
        else                      rgbPurple();
        lastColorChange = millis();
    }
}

//        BUZZER HELPERS
```

```
void buzzerAlternate() {
    if (millis() - lastBuzzerChange >= 1000) {
        buzzerPhase = (buzzerPhase + 1) % 2;
        lastBuzzerChange = millis();
        if (buzzerPhase == 0) { tone(BUZZER1,4000); noTone(BUZZER2); }
        else                  { tone(BUZZER2,4000); noTone(BUZZER1); }
    }
}
void buzzerOff() {
    noTone(BUZZER1); noTone(BUZZER2);
    digitalWrite(BUZZER1,LOW); digitalWrite(BUZZER2,LOW);
    buzzerPhase = 0; lastBuzzerChange = 0;
}
void buzzerBeep(int ms) {
    tone(BUZZER1,4000); delay(ms);
    noTone(BUZZER1); digitalWrite(BUZZER1,LOW);
}

//        ULTRASONIC

long getDistance() {
    digitalWrite(TRIG_PIN,LOW);  delayMicroseconds(2);
    digitalWrite(TRIG_PIN,HIGH); delayMicroseconds(10);
    digitalWrite(TRIG_PIN,LOW);
    long dur = pulseIn(ECHO_PIN, HIGH, 30000);
    return dur * 0.034 / 2;
}

//        LCD HELPER

void lcdPrint(int row, const char* msg) {
    lcd.setCursor(0, row);
    lcd.print("                    ");
    lcd.setCursor(0, row);
    lcd.print(msg);
}

//        RESET

void resetSystem() {
    wrongAttempts = 0; enteredPass = "";
    systemArmed = false; colorPhase = 0;
    buzzerOff(); rgbOff(); lcd.clear(); lcd.noBacklight();
}

//        I2C SCAN

byte scanI2C() {
    for (byte addr = 1; addr < 127; addr++) {
        Wire.beginTransmission(addr);
        if (Wire.endTransmission() == 0) return addr;
    }
    return 0;
```

```
}

//        SETUP


void setup() {
    pinMode(BUZZER1,OUTPUT); pinMode(BUZZER2,OUTPUT);
    pinMode(RGB_RED,OUTPUT); pinMode(RGB_GREEN,OUTPUT);
    pinMode(RGB_BLUE,OUTPUT);
    pinMode(TRIG_PIN,OUTPUT); pinMode(ECHO_PIN,INPUT);
    pinMode(PIR_PIN,INPUT);
    buzzerOff(); rgbOff();
    Serial.begin(9600); Wire.begin();
    lcdAddress = scanI2C();
    if (lcdAddress != 0) {
        lcd = LiquidCrystal_I2C(lcdAddress, 16, 2);
        lcd.init(); lcd.noBacklight();
    }
}

//        LOOP


void loop() {
    if (!systemArmed) {
        long dist = getDistance();
        bool pirTriggered = digitalRead(PIR_PIN) == HIGH;
        if ((dist > 0 && dist <= 30) || pirTriggered) {
            systemArmed = true; enteredPass = "";
            wrongAttempts = 0; buzzerPhase = 0;
            rgbBlue(); lcd.backlight(); lcd.clear();
            lcdPrint(0, "  Enter PIN:    ");
        }
        return;
    }

    bool pirActive = digitalRead(PIR_PIN) == HIGH;

    if (wrongAttempts >= 2 || pirActive) {
        if (pirActive) {
            buzzerAlternate(); cycleRGB();
            lcdPrint(0, "  ALARM!!!      ");
        } else {
            buzzerOff();
            if (millis() - lastColorChange >= 300) {
                colorPhase = (colorPhase + 1) % 2;
                if (colorPhase == 0) rgbRed(); else rgbOff();
                lastColorChange = millis();
            }
            lcdPrint(0, "  Enter PIN:    ");
        }
        char key = keypad.getKey();
        if (key) {
            if (key == '*') { enteredPass = ""; lcdPrint(1,"
        "); }
            else if (key != 'A' && key != 'B' && key != 'C'
                     && key != 'D' && key != '#') {
                enteredPass += key;
```

```
                lcd.setCursor(0,1); lcd.print(enteredPass);
                if (enteredPass.length() == 4) {
                    if (enteredPass == PASSWORD) {
                        buzzerOff(); rgbGreen(); lcd.clear();
                        lcdPrint(0,"    Accepted!   "); delay(2000);
                        resetSystem();
                    } else {
                        wrongAttempts++; enteredPass = "";
                        lcdPrint(1,"  Wrong PIN!    "); delay(1000);
                        lcdPrint(1,"                ");
                    }
                }
            }
        }
        return;
    }

    if (wrongAttempts == 1) rgbRed();
    char key = keypad.getKey();
    if (!key) return;
    if (key == 'A' || key == 'B' || key == 'C' || key == 'D' || key == '#') return;
    if (key == '*') { enteredPass = ""; lcdPrint(1,"                "); return; }
    enteredPass += key;
    lcd.setCursor(0,1); lcd.print(enteredPass);
    if (enteredPass.length() == 4) {
        if (enteredPass == PASSWORD) {
            buzzerOff(); rgbGreen(); lcd.clear();
            lcdPrint(0,"    Accepted!   "); delay(2000); resetSystem();
        } else {
            wrongAttempts++; enteredPass = "";
            lcdPrint(1,"  Wrong PIN!    ");
            if (wrongAttempts == 1) { buzzerBeep(500); rgbRed(); }
            delay(1000); lcdPrint(1,"                ");
        }
    }
}
```

Listing 6: Complete Arduino firmware – Home Intrusion Detection System

6 Results and Discussion

6.1 System Behaviour Verification

The system was tested against each defined state transition. Table 3 summarises the test scenarios and observed outcomes.

Table 3: Test Scenarios and Observed Outcomes

Test	Stimulus	Expected Output	Result
T1	Object within 30 cm of ultrasonic sensor	LCD on, blue LED, PIN prompt	Pass
T2	PIR detects motion in IDLE	LCD on, blue LED, PIN prompt	Pass
T3	Correct PIN (1234) entered	Green LED, "Accepted!", reset	Pass
T4	Incorrect PIN (1 attempt)	Red LED, beep, re-prompt	Pass
T5	Incorrect PIN (2 attempts)	Red LED flashing, re-prompt	Pass
T6	PIR active after wrong attempt	Dual buzzer, cycling LED, "ALARM!!!"	Pass
T7	Correct PIN during ALARM	All outputs off, system reset	Pass
T8	'*' key pressed	PIN buffer cleared	Pass
T9	I^2C LCD not at 0x27	Dynamic address resolved at startup	Pass

6.2 Non-Blocking Timing

A key design decision was the use of `millis()`-based timing for LED cycling and buzzer alternation instead of `delay()`. This ensures that keypad polling remains responsive during the ALARM state. Measurements confirmed that key presses registered within one `loop()` iteration (<1 ms) regardless of the 200 ms LED or 1000 ms buzzer intervals.

6.3 Sensor Fusion Effectiveness

The combination of the ultrasonic and PIR sensors addressed two common failure modes:

- **Ultrasonic alone:** Susceptible to false negatives when the target approaches from the side (outside the 15° beam angle).
- **PIR alone:** Susceptible to false negatives when the subject moves very slowly or is thermally similar to the environment.

Using both sensors in a logical OR configuration ensured consistent triggering across all tested approach scenarios.

6.4 Limitations

1. **Hard-coded PIN:** The password `"1234"` is stored as a compile-time constant. A production system should use EEPROM-backed runtime PIN management.
2. **Single-factor authentication:** PIN entry alone is vulnerable to shoulder-surfing. An additional factor (e.g., RFID card) would strengthen security.
3. **No network connectivity:** The system provides only local alarm output. Remote notification (SMS, Wi-Fi push) would be a valuable extension.
4. **Ultrasonic blind spot:** The HC-SR04 has a minimum measurable range of approximately 2 cm; targets closer than this return no echo.

7 Conclusion

This project successfully designed and implemented a functional home intrusion detection system on the Arduino Mega 2560 platform. The system demonstrates a well-structured finite-state machine with five operational states, reliable dual-sensor triggering, PIN-based authentication, and non-blocking multi-modal output control. All nine test scenarios passed, confirming correct state transitions and output behaviour.

The project reinforced several core microcontroller engineering principles: digital I/O management, I^2C serial communication, non-blocking real-time loop design, and sensor fusion. Future work should address the identified limitations by adding EEPROM PIN storage, RFID-based second-factor authentication, and IoT connectivity for remote alerting.

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